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Compiler vs Interpreter

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Course name: Compiler Design

Course Code: CSA1407



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What is a Compiler?

- Translates entire source code at once
- Produces executable file
- Errors shown after compilation
- Faster execution



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What is an Interpreter?

- Translates line by line
- No separate executable
- Stops at first error
- Slower execution



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Key Differences

- Compilation: Whole program vs Line by line
- Output: Executable vs Immediate execution
- Speed: Faster vs Slower
- Errors: After compile vs During run



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Advantages

- Compiler: Better performance, optimization
- Interpreter: Easier debugging, portability



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Examples

- Compiled: C, C++, Go, Rust
- Interpreted: Python, JavaScript, Ruby
- Hybrid: Java (bytecode + JVM)



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Conclusion

- Choose a compiler for speed and deployment.
- Choose an interpreter for rapid development and testing.
- Thank You!

THANK YOU